Function Notation: $f(x)$: When we write an equation $y = f(x)$ by $f(x)$ we mean a rule which processes the value of variable x and assigns it to variable y. Thus $f(x)$ is essentially a mapping rule which outputs a value of y corresponding to a given value of x. For $f(x)$ to qualify as a function, the output value of y must be unique.

Domain And Range Of A Function: The set of all possible allowable values of x for a function is called domain of the function and the set of all possible output values is called Range of the function. As an example, domain and range of some functions is listed in the table below:

Function	Domain (x)	Range (y)
$y = x^2$	$(-\infty, \infty)$	$[0, \infty)$
$y = 1/x$	$(-\infty, 0) \cup (0, \infty)$	$(-\infty, 0) \cup (0, \infty)$
$y = \sqrt{x}$	$[0, \infty)$	$[0, \infty)$
$y = \sqrt{4 - x}$	$(-\infty, 4]$	$[0, \infty)$
$y = \sqrt{1 - x^2}$	$[-1, 1]$	$[0, 1]$

Think: How can one arrive at these domains and ranges.

Even Function: A function is said to be even if $f(-x) = f(x)$ for example x^2, $\cos(x)$, $1/x^2$ etc. The graph of every even function is symmetric about y axis.

Odd Function: A function is said to be odd if $f(-x) = -f(x)$. Every odd function is symmetric about the origin, which means that graph remains unchanged upon 180° rotation about the origin (on the portion in second and third quadrant is water image of the mirror image of portion in I and IV Quadrants.

Example: If $f(x) = \sqrt{x}$ and $g(x) = x + 1$ find:
(a) $f(g(x))$ (b) $g(f(x))$ (c) $f(f(x))$ (d) $g(g(x))$

YOUTUBE.COM/@THEPHYSICSSHELTER

Solution: (a) $f(g(x)) = f(x+1) = \sqrt{x+1}$ Ans.

(b) $g(f(x)) = g(\sqrt{x}) = \sqrt{x}+1$ Ans.

(c) $f(f(x)) = f(\sqrt{x}) = \sqrt{\sqrt{x}} = x^{1/4}$ Ans.

(d) $g(g(x)) = g(x+1) = x+2$ Ans.

EXAMPLE: What is the domain and range of each function that you obtained in parts a, b, c and d above?

Solution: (a) $\sqrt{x+1}$ is defined for $x+1 \geq 0 \Rightarrow$ D: $x \in [-1, \infty)$ its value can be anywhere from 0 to ∞ thus: R: $y \in [0, \infty)$

(b) $\sqrt{x}+1$ is defined for $x \geq 0$ thus D: $x \in [0, \infty]$, its value can be anywhere between 1 and ∞ thus R: $y \in [1, \infty)$

(c) $x^{1/4}$ is defined for $x > 0$ thus D: $x \in [0, \infty)$ its value can be anywhere between 0 and ∞ thus R: $y \in [0, \infty)$

(d) $x+2$ is defined for all values of x thus D: $x \in (-\infty, \infty)$ it can take all possible values from $-\infty$ to ∞ thus R: $y \in (-\infty, \infty)$.

MEANING OF INSTANTANEOUS RATE OF CHANGE: (RICHARD FEYNMAN)

A lady in a car is pulled over by a cop, and the cop says, "Lady, you were going 80 Km per hour, so you will be challaned for over speeding, the permissible speed limit is 50Km/hr!" She says, "That's impossible, sir, I was travelling for only seven minutes. It is ridiculous—how can I go 80Km/hr when I haven't gone for an hour?" How would you answer her if you were the cop? The Cop said "If you keep going like this, you would cover 80 Km in 1 hour, and then she replied, "How do you know that I would keep going like this, my foot is off the accelerator pedal, my car would come to a stop after a while, or maybe I would bump in to the wall at the end of the road". The cop got thoroughly confused, but let us try to understand, what we mean by the idea that speed right now is 80Km/hr. "Right now,", is just an instant which is nothing but an interface between past and future so literally an interval of zero time. Obviously in a literal zero time, the distance travelled will also be zero, and then our instantaneous speed is 0/0 ... Lol! 😊.

YOUTUBE.COM/@THEPHYSICSSHELTER

Newton was very much bothered by this kind of 0/0 forms involved in discussing instantaneous phenomena... and this led to his invention of differential calculus. We cannot find 0/0 however we can find the ratios of finite quantities no matter how so small. So, we can find the distance travelled by the lady in 1 second and divide it by 1 second to get average speed for 1 second. We can similarly reduce the time interval further to 0.1 second and find the average speed for 0.1 second, we can also make this time interval successively smaller to 0.01 second or 0.0001 second or anything arbitrarily close to zero so long as it is not literal zero, the ratio of distance to time will be defined. This ratio when the time interval is arbitrarily close to zero is called the instantaneous speed, and the distance and time involved here are said to be infinitesimal i.e., arbitrarily close to zero (as close to zero as you can imagine it to be) but not literal zero.

YOUTUBE.COM/@THEPHYSICSSHELTER

Average And Instantaneous Speeds:

Example: When a stone falls from rest in vacuum, it follows Galileo's law i.e its distance travelled till a time t is given as: $y = 4.9t^2$ (essentially $\frac{1}{2}gt^2$). Here y is in metres and t is in seconds. Consider a general time interval between t and $t+\Delta t$. (a) What is the average speed of the stone between the general time duration extending from t to $t+\Delta t$. (btw Δ just means "change in". So if time changes from 1 sec to 1.2 sec, $\Delta t = 0.2$ seconds). (b) Evaluate the average speed of the stone after $t = 1$ sec for intervals of time $\Delta t = 1$ sec, 0.1 sec, 0.01 sec, 0.001 sec and 0.0001 sec (c) Evaluate average speed of stone after $t = 2$ sec for intervals of time $\Delta t = 1$ sec, 0.1 sec, 0.01 sec, 0.001 sec and 0.0001 sec (d) Do you spot a pattern, [what happens as Δt becomes arbitrarily small?)

(e) What are the instantaneous speeds at $t = 1$ sec and $t = 2$ sec? and what is the instantaneous speed at a general instant of time t?

Solution: (a) $y = f(t) = 4.9t^2$, now average speed for any interval of time is simply $\Delta y/\Delta t$ (Δy is "change in y" for the interval of time Δt). Thus:

$\Delta y = f(t+\Delta t) - f(t)$ i.e $4.9(t+\Delta t)^2 - 4.9t^2$ i.e:

$\Delta y = (2t\Delta t + (\Delta t)^2)4.9 \Rightarrow \quad \Delta y/\Delta t = 4.9(2t + \Delta t)$ Ans.

(b) Now we just need to put $t = 1$ sec and put various values of Δt to get the desired average speeds which will be presented shortly along with results for part (c).

(c) Here we just need to put $t = 2$ sec and $\Delta t = 1, 0.1, 0.01, 0.001$ and 0.0001 sec to get the desired results. The calculations for parts (b) and (c) are presented in the table below:

Average speeds over short time intervals $[t, t+\Delta t]$

$$\text{Average speed: } \frac{\Delta y}{\Delta t} = \frac{4.9(t+\Delta t)^2 - 4.9t^2}{\Delta t}$$

Length of time interval (Δt)	Average speed over interval of length Δt starting at $t = 1$	Average speed over interval of length Δt starting at $t = 2$
1	14.7	24.5
0.1	10.29	20.09
0.01	9.849	19.649
0.001	9.8049	19.6049
0.0001	9.80049	19.60049

(d) From above table one can clearly see that as the intervals are becoming smaller and smaller, at $t = 1$ sec the speed is becoming closer and closer to $9.8\ ms^{-1}$ and similarly at $t = 2$ sec it is becoming closer and closer to $19.6\ ms^{-1}$, as Δt becomes arbitrarily small we expect the speeds to become $9.8\ ms^{-1}$ and $19.6\ ms^{-1}$ respectively.

(e) In the light of our above discussion we can say that our instantaneous speed at any instant of time is given as:

$$\lim_{\Delta t \to 0} \frac{4.9(t+\Delta t)^2 - 4.9t^2}{\Delta t}$$

Here the notation $\lim_{\Delta t \to 0}$ is read as "Limit Delta t

YOUTUBE.COM/@THEPHYSICSSHELTER

tending to zero" and it means that we are evaluating the expression for a value of Δt being arbitrarily close to zero (or infinitesimal value of Δt). Thus instantaneous speed becomes

$$\lim_{\Delta t \to 0} \frac{4.9\left[t^2 + 2t\Delta t + (\Delta t)^2 - t^2\right]}{\Delta t} = \lim_{\Delta t \to 0} \frac{4.9(2t\Delta t + \Delta t^2)}{\Delta t}$$

$$= \lim_{\Delta t \to 0} 4.9(2t + \Delta t)$$ Now think what happens when Δt is arbitrarily close to zero, in our last expression we will not face any problem of division by zero even if we put Δt equal to zero, thus our instantaneous speed is simply $4.9 \times 2t$ i.e $v = 9.8t$ ans. Now for $t = 1$, $v = 9.8\ ms^{-1}$ and for $t = 2$, $v = 19.6\ ms^{-1}$ ans.

Derivative (or Instantaneous Rate Of Change): Let $y = f(t)$ be some function of time the we define instantaneous rate of change of y with respect to t as:

$$\frac{dy}{dt} = \lim_{\Delta t \to 0} \frac{f(t+\Delta t) - f(t)}{\Delta t} \quad (1)$$

Here dy is read as infinitesimal change in y and dt is similarly infinitesimal change in t. The quantity dy/dt is read as:

1. Derivative of y with respect to t
2. Differentiation of y with respect to t
3. Instantaneous rate of change of y wrt t.

Now using definition of equation (1) above, we can evaluate the differentiation of any function. We shall show the differentiation of some simple functions here and then just list out the table for derivatives of standard functions. (Deriving some of these results will require a still deeper understanding and practice. All these results will be derived from basics when you study the chapter on Limit continuity and differentiability during your JEE Mathematics

YOUTUBE.COM/@THEPHYSICSSHELTER

course).

EXAMPLE: Find the derivative of the function $y = f(t) = t^3$ using the definition.

Solution: $\frac{dy}{dt} = \lim_{\Delta t \to 0} \frac{f(t+\Delta t) - f(t)}{\Delta t} =$

$= \lim_{\Delta t \to 0} \frac{(t+\Delta t)^3 - t^3}{\Delta t} = \lim_{\Delta t \to 0} \frac{t^3 + 3t^2\Delta t + 3t(\Delta t)^2 + (\Delta t)^3 - t^3}{\Delta t}$

$= \lim_{\Delta t \to 0} 3t^2 + 3t\Delta t + (\Delta t)^2$, Now we can set Δt to zero

Thus $\frac{dy}{dt} = 3t^2$.

EXAMPLE: Find derivative of $y = f(t) = t^n$ using the definition and binomial theorem.

Solution: $\frac{dy}{dt} = \lim_{\Delta t \to 0} \frac{(t+\Delta t)^n - t^n}{\Delta t} = \lim_{\Delta t \to 0} \frac{t^n[(1+\Delta t/t)^n - 1]}{\Delta t}$

$\lim_{\Delta t \to 0} \frac{t^n}{\Delta t}\left[1 + n\frac{\Delta t}{t} + \frac{n(n-1)}{2}\left(\frac{\Delta t}{t}\right)^2 + \dots - 1\right]$

$= \lim_{\Delta t \to 0} n t^{n-1} + \frac{n(n-1)}{2} t^{n-2}\Delta t + \dots$ Now setting $\Delta t \to 0$, we get

$\frac{dy}{dt} = n t^{n-1}$

EXAMPLE: Using first principles find the derivative of $y = f(t) = \sin(t)$.

Solution: $\frac{dy}{dt} = \lim_{\Delta t \to 0} \frac{\sin(t+\Delta t) - \sin(t)}{\Delta t} = \lim_{\Delta t \to 0} \frac{2\cos(t+\Delta t/2)\sin(\Delta t/2)}{\Delta t}$

Now for very small angles recall that, $\left(\because \sin C - \sin D = 2\cos\left(\frac{C+D}{2}\right)\sin\left(\frac{C-D}{2}\right)\right)$

$\sin\theta \approx \theta$ thus $\sin(\Delta t/2) \approx (\Delta t/2)$

when Δt is very small, thus, $\frac{dy}{dt} = \lim_{\Delta t \to 0} \frac{2\cos(t+\Delta t/2)\cdot(\Delta t/2)}{\Delta t}$

$\Rightarrow \frac{dy}{dt} = \lim_{\Delta t \to 0} \cos(t+\Delta t/2) = \underline{\cos(t)}$

EXAMPLE: Using first principles find the derivative of $y = f(t) = \ln(t)$.

Solution: $\frac{dy}{dt} = \lim_{\Delta t \to 0} \frac{\ln(t+\Delta t) - \ln(t)}{\Delta t} = \lim_{\Delta t \to 0} \frac{1}{\Delta t}\ln\left(\frac{t+\Delta t}{t}\right)$

YOUTUBE.COM/@THEPHYSICSSHELTER

$$= \lim_{\Delta t \to 0} \frac{1}{\Delta t} \ln\left(1 + \frac{\Delta t}{t}\right) = \lim_{\Delta t \to 0} \frac{1}{\Delta t}\left[\frac{\Delta t}{t} - \frac{1}{2}\left(\frac{\Delta t}{t}\right)^2 + \frac{1}{3}\left(\frac{\Delta t}{t}\right)^3 + \cdots\right]$$

$$= \lim_{\Delta t \to 0}\left[\frac{1}{t} - \frac{1}{2}\frac{\Delta t}{t^2} + \frac{1}{3}\frac{\Delta t^2}{t^3} + \cdots\right] \quad \because \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3}, \cdots$$

Now setting $\Delta t = 0$. Thus $dy/dt = 1/t$ Ans.

EXAMPLE: If $y = f(x) = e^x$, find dy/dx using basic definition.

Solution: $\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x} = \lim_{\Delta x \to 0} \frac{e^{x+\Delta x} - e^x}{\Delta x}$

$$= \lim_{\Delta x \to 0} \frac{e^x}{\Delta x}\left[e^{\Delta x} - 1\right] = \lim_{\Delta x \to 0} \frac{e^x}{\Delta x}\left[1 + \Delta x + \frac{(\Delta x)^2}{2!} + \frac{(\Delta x)^3}{3!} + \cdots - 1\right]$$

$$= \lim_{\Delta x \to 0} e^x\left[1 + \frac{\Delta x}{2!} + \frac{(\Delta x)^2}{3!} + \cdots\right] = e^x \text{ Ans.} \quad \left(\because e^x = 1 + x + \frac{x^2}{2!} + \cdots\right)$$

DERIVATIVE OF STANDARD FUNCTIONS :

	f(x)	f'(x)
(i)	x^n	nx^{n-1}
(ii)	e^x	e^x
(iii)	a^x	$a^x \ln(a)$, $a > 0$
(iv)	ℓnx	$1/x$
(v)	$\log_a x$	$(1/x)\log_a e$, $a > 0$, $a \neq 1$
(vi)	sinx	cosx
(vii)	cosx	$-$ sinx
(viii)	tanx	$\sec^2 x$
(ix)	secx	secx tanx

YOUTUBE.COM/@THEPHYSICSSHELTER

(x)	$\text{cosec}x$	$-\text{cosec}x \cdot \cot x$
(xi)	$\cot x$	$-\text{cosec}^2 x$
(xii)	constant	0
(xiii)	$\sin^{-1} x$	$\frac{1}{\sqrt{1-x^2}}, -1 < x < 1$
(xiv)	$\cos^{-1} x$	$\frac{-1}{\sqrt{1-x^2}}, -1 < x < 1$
(xv)	$\tan^{-1} x$	$\frac{1}{1+x^2}, \ x \in R$
(xvi)	$\sec^{-1} x$	$\frac{1}{\lvert x \rvert \sqrt{x^2-1}}, \lvert x \rvert > 1$
(xvii)	$\text{cosec}^{-1} x$	$\frac{-1}{\lvert x \rvert \sqrt{x^2-1}}, \lvert x \rvert > 1$
(xviii)	$\cot^{-1} x$	$\frac{-1}{1+x^2}, x \in R$

YOUTUBE.COM/@THEPHYSICSSHELTER

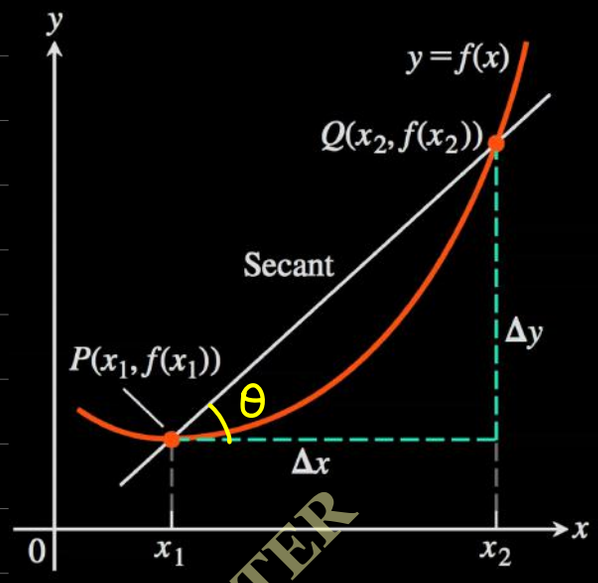
→ If $y = f(x)$ then y' or $f'(x)$ mean dy/dx

→ If $y = f(t)$ then $\dot{y}$ or $\dot{f}(t)$ represents dy/dt.

Average Rate Of Change And Secant Line:

If $y = f(x)$ is a function of x then between two coordinates x_1 and x_2 the average rate of change of y wrt x is defined as ratio of total change in y to total change in x. i.e,

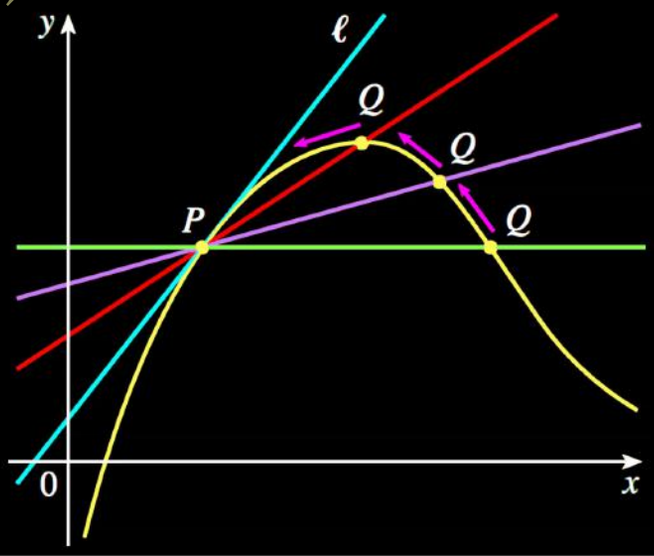
$$\text{Average rate of change} = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$



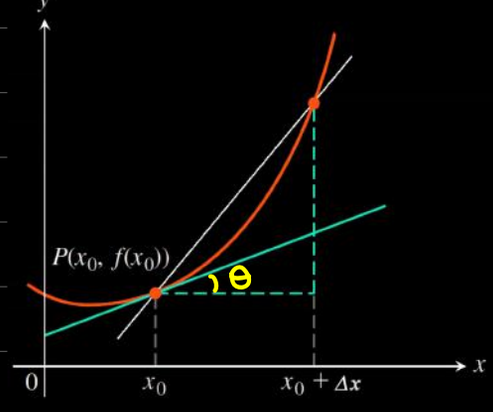
Refer to the adjacent figure, we can easily infer that average rate of change is simply $\Delta y/\Delta x = \tan\theta$ where θ is the angle of the secant between points (x_2, y_2) and (x_1, y_1) with the x-axis. $(y_1 = f(x_1),\ y_2 = f(x_2))$

Instantaneous Rate Of Change And Tangent:

In the previous section we saw how average rate of change is related to the secant of the curve and earlier we saw that instaneous rate of change is nothing but average rate of change for an infinitesimal time interval (or space interval). This is equivalent to finding the average rate of change by bringing Q arbitrarily close to P (see adjacent figure). However as Q is brought closer and closer to P, it tends to become the tangent and in the infinitesimal can be treated as the tangent itself.



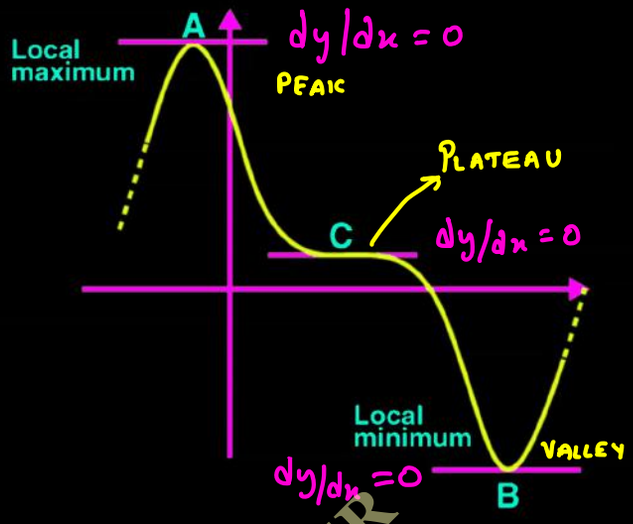
Thus the derivative of a function is represented by: $\frac{dy}{dx} = \tan\theta$ where θ is the angle which the tangent of the curve makes with the positive x axis (upward tangent to be precise). The value of $\tan\theta$ here is also known as slope of the curve at point P



YOUTUBE.COM/@THEPHYSICSSHELTER

(See the figure).

→ One can easily see that wherever there is a peak or a valley in a smooth function, the tangent becomes parallel to x axis thus tanθ becomes zero. This gives us a technique to locate such peaks and valleys in the graph in a function. (Here $dy/dx = \tan\theta = 0$) We shall explore this idea in greater detail in our section of maxima and minima).



Rules For Differentiation:

Constant Multiple Rule:

$$\frac{d}{dx}[c\,f(x)] = c\,\frac{d}{dx}(f(x))$$

i.e. a multiplied constant can be taken outside the differentiation operator. For instance:

$$\frac{d}{dx}(3x^4) = 3\,\frac{d}{dx}(x^4) = 3 \times 4x^3 = 12x^3$$

Sum And Difference Rules:

$$\frac{d}{dx}[f(x) + g(x)] = \frac{d}{dx}[f(x)] + \frac{d}{dx}[g(x)]$$

$$\frac{d}{dx}[f(x) - g(x)] = \frac{d}{dx}[f(x)] - \frac{d}{dx}[g(x)]$$

For instance, $\frac{d}{dx}(x^8 + 12x^5 - 4x^4 + 10x^3 - 6x + 5)$

$$\frac{d}{dx}x^8 + 12\frac{d}{dx}x^5 - 4\frac{d}{dx}x^4 + 10\frac{d}{dx}x^3 - 6\frac{d}{dx}(x) + \frac{d}{dx}(5)$$

$$= 8x^7 + 60x^4 - 16x^3 + 30x^2 - 6 \quad \text{Ans.}$$

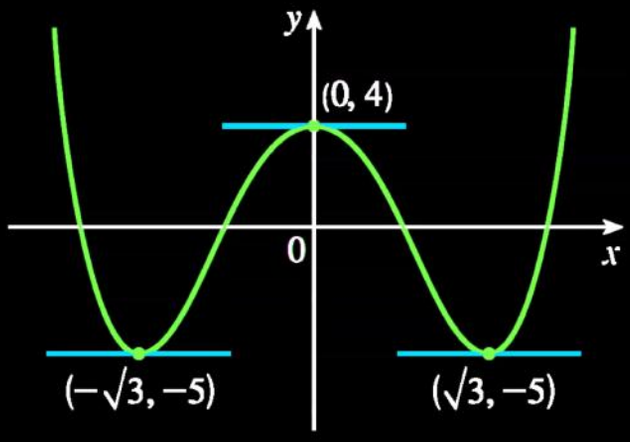
YOUTUBE.COM/@THEPHYSICSSHELTER

EXAMPLE: Find the points on the curve $y = x^4 - 6x^2 + 4$ where the tangent line is horizontal.

Solution: At the points of horizontal tangents, derivative of the function must be zero i.e, $\frac{d}{dx}(x^4 - 6x^2 + 4) = 0$

$\Rightarrow 4x^3 - 12x = 0 \Rightarrow 4x(x^2 - 3) = 0$

$\Rightarrow x = 0, \sqrt{3}$ and $-\sqrt{3}$, These points can be identified in the adjacent graph.



EXAMPLE: The equation of motion of a particle is $s = 2t^3 - 5t^2 + 3t + 4$ where s is in cm and t is in seconds. Find the acceleration as a function of time and its value at $t = 2$ sec. (Note that acceleration is derivative of velocity which in turn is derivative of position or displacement).

Solution: velocity $v = \frac{ds}{dt} = 6t^2 - 10t + 3$

acceleration $= a = dv/dt = 12t - 10$ Ans.

Thus $a(2) = 12 \times 2 - 10 = 14\ cm\ s^{-2}$ Ans.

→ Note that derivative of the derivative is also known as second derivative, the notation used for it is d^2y/dx^2 or y'' or $f''(x)$. In case of second time derivative we write the derivative as d^2y/dt^2 or $\ddot{y}$ or $\ddot{f}(t)$.

→ The general notation d^ny/dx^n represents n^{th} derivative (i.e. carrying out the differentiation process successively n times).

→ The notations like y'' or $\ddot{y}$ are convenient for writing small order derivatives and were introduced by Gottfried Leibniz (a french mathematician who worked independent of Newton for developing calculus). Infact there was a controversy between Leibniz and Newton regarding claims to inventing ideas in calculus, Newton's supporters accuse Leibniz of plagiarizing Newton's unpublished

YOUTUBE.COM/@THEPHYSICSSHELTER
AMBARISH SIR
https://www.youtube.com/@ambarishphysics https://www.youtube.com/@thephysicsshelter

ideas, however modern consensus is that both of them developed their ideas independently.

PRODUCT RULE: $\boxed{\frac{d}{dx}\left[f(x)\,g(x)\right] = f(x)\frac{d}{dx}[g(x)] + g(x)\frac{d}{dx}[f(x)]}$

EXAMPLE: If $f(x) = xe^x$ find (a) $f'(x)$ (b) n^{th} derivative $f^{(n)}(x)$.

Solution: $f'(x) = \frac{d}{dx}(xe^x) = x\frac{d}{dx}(e^x) + e^x\cdot\frac{d}{dx}(x)$

$= xe^x + e^x = e^x(x+1)$ Ans.

We can see that $f'(x) = f(x) + e^x$

Thus $f''(x) = f'(x) + e^x = 2f(x) + e^x$ i.e $2xe^x + e^x$

Similarly $f'''(x) = f''(x) + e^x =$

$\vdots \qquad \vdots$

$f^n(x) = f^{(n-1)}(x) + e^x$

$f^n(x) = f(x) + xe^x = e^x(x+n)$ Ans.

EXAMPLE: Differentiate the function $f(t) = \sqrt{t}(a+bt)$ wrt t.

Solution: Method 1: $\frac{df}{dt} = \sqrt{t}\frac{d}{dt}(a+bt) + (a+bt)\cdot\frac{d}{dt}(\sqrt{t})$

$\Rightarrow \frac{df}{dt} = b\sqrt{t} + (a+bt)\cdot\frac{1}{2\sqrt{t}} = \frac{a+3bt}{2\sqrt{t}}$ Ans.

Method 2: $f(t) = a\sqrt{t} + bt^{3/2} \Rightarrow \frac{df}{dt} = \frac{a}{2\sqrt{t}} + \frac{3}{2}bt^{1/2}$

$\Rightarrow \frac{df}{dt} = \frac{a+3bt}{2\sqrt{t}}$ Ans.

EXAMPLE: $f(x) = \sqrt{x}\,g(x)$, where $g(4) = 2$ and $g'(4) = 3$ find $f'(4)$.

Solution: $f'(x) = \sqrt{x}\,g'(x) + g(x)\cdot\frac{1}{2\sqrt{x}}$

$\Rightarrow f'(4) = \sqrt{4}\times g'(4) + g(4)\cdot\frac{1}{2\sqrt{4}} = 6 + \frac{1}{2} = 6.5$ Ans.

YOUTUBE.COM/@THEPHYSICSSHELTER
AMBARISH SIR
https://www.youtube.com/@ambarishphysics
https://www.youtube.com/@thephysicsshelter

Quotient Rule:

$$\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g(x)\, d/dx[f(x)] - f(x)\, d/dx(g(x))}{[g(x)]^2}$$

Example: Let $y = (x^2+x-2)/(x^3+6)$ find dy/dx.

Solution: $dy/dx = \dfrac{(x^3+6)\, d/dx(x^2+x-2) - (x^2+x-2)\, d/dx(x^3+6)}{(x^3+6)^2}$

$$= \frac{(x^3+6)(2x+1) - (x^2+x-2)(3x^2)}{(x^3+6)^2} = \frac{(2x^4+x^3+12x+6) - (3x^4+3x^3-6x^2)}{(x^3+6)^2}$$

$$= \frac{-x^4 - 2x^3 + 6x^2 + 12x + 6}{(x^3+6)^2} \text{ ans.}$$

→ Don't use quotient rule just because something is written as a quotient, sometimes it is easier to perform the actual division before differentiating. e.g. If $F(x) = (3x^2 + 2\sqrt{x})/x$ instead of using quotient rule to find $F'(x)$ we can first write $F(x) = 3x + 2x^{-1/2}$ Thus $F'(x) = 3 - x^{-3/2}$ ans.

Chain Rule: Suppose we have a function of the form $F(x) = \sqrt{x^2+1}$, this function is not of any standard form that we had listed in our table. However if we let $u = x^2+1$ then our function is $y = F(x) = \sqrt{u}$, Now we find $\dfrac{dy}{dx} = \dfrac{dy}{du} \times \dfrac{du}{dx}$.

Now $\dfrac{dy}{du} = \dfrac{1}{2\sqrt{u}}$ and $\dfrac{du}{dx} = 2x$ thus required derivative is:

$$\frac{dy}{dx} = \frac{1}{2\sqrt{x^2+1}} \cdot 2x = \frac{x}{\sqrt{x^2+1}} \text{ ans.}$$

Thus Whenever we have a function of a function, $y = f(g(x))$ then $y' = f'(g(x)) \times g'(x)$ where $f'(g(x))$ represents derivative of f with respect to $g(x)$ and $g'(x)$ represents derivative of

YOUTUBE.COM/@THEPHYSICSSHELTER
AMBARISH SIR
https://www.youtube.com/@ambarishphysics
https://www.youtube.com/@thephysicsshelter

g(x) wrt x. In Leibniz notation:

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} \quad \text{or} \quad y' = F'(g(x)) \cdot g'(x)$$

For applying chain rule, we can start working from outermost function without bothering about its actual argument and then keep working inwards. What we mean by this will be clear from following example:

Suppose $y = \sqrt{x^2+1}$, Here outermost function is $\sqrt{\text{something}}$

we are initially not worried what actually that "something" is, because for the first term we are just taking the derivative wrt that "something" itself thus for first term we we write $\frac{1}{2\sqrt{\text{something}}}$,

just as the derivative of $\sqrt{x}$ wrt x is $1/2\sqrt{x}$. Thus our first term becomes $\frac{1}{2\sqrt{x^2+1}}$ and now for second term we need to

differentiate this "something" wrt x, which is $2x$, thus our required derivative becomes $\frac{1}{2\sqrt{x^2+1}} \cdot 2x = x/\sqrt{x^2+1}$ Ans.

We could use this method repeatedly for multiple nestings as well for instance. If $y = f(g(h(x)))$ then $\frac{dy}{dx} = \frac{dy}{dg} \times \frac{dg}{dh} \times \frac{dh}{dx}$, Here,

while writing $\frac{dy}{dg}$ we need not worry about h, Just deal with the block

g as "something". This will be clearer from following example.

Suppose $y = \sin(\cos(\tan(x)))$, Here $f = \sin$, $g = \cos$, $h = \tan$

Thus we write $y' = \cos(\underbrace{\cos\tan(x)}_{\text{something}}) \times -\sin(\underbrace{\tan(x)}_{\text{something}}) \times \sec^2(x)$

Sin (something) was just replaced by Cos (something) and so on.

AMBARISH SIR

https://www.youtube.com/@ambarishphysics https://www.youtube.com/@thephysicsshelter

YOUTUBE.COM/@THEPHYSICSSHELTER

EXAMPLE: Differentiate $y = (x^3-1)^{100}$ wrt x.

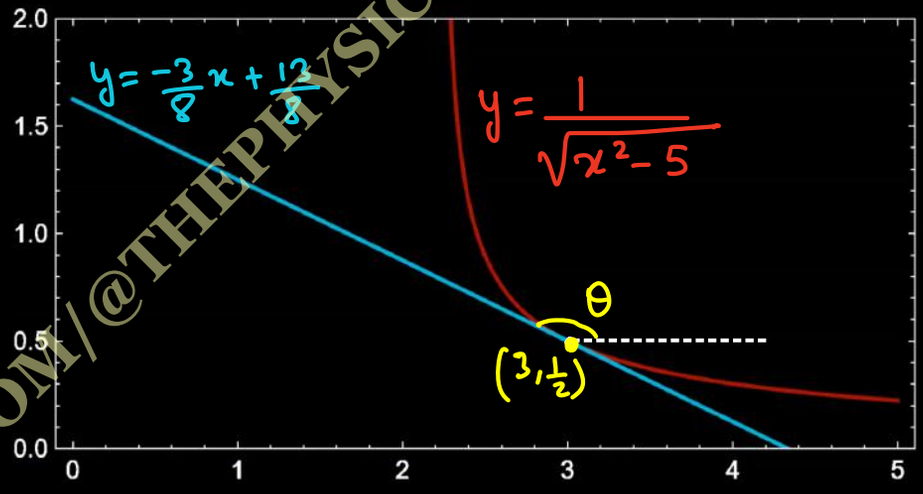
Solution: Using chain rule, $y' = 100(x^3-1)^{99} \times 3x^2$ Ans.

EXAMPLE: Differentiate $y = 1/\sqrt[3]{x^2+x+1}$ wrt x.

Solution: $y = (x^2+x+1)^{-1/3}$, Using chain rule,

$$\frac{dy}{dx} = \frac{-1}{3}(x^2+x+1)^{-4/3} \times (2x+1) \text{ Ans.}$$

EXAMPLE: Suppose $y = f(x) = 1/\sqrt{x^2-5}$ find the equation of tangent to $f(x)$ at $(3, 1/2)$.



Solution: Recall that equation of straight line is given by $y = mx + c$ where m is slope of the line. For tangent line, slope of the tangent is same as slope of the curve thus:

$$m = \frac{dy}{dx}, \text{ Now } y = \frac{1}{\sqrt{x^2-5}} = (x^2-5)^{-1/2} \Rightarrow \frac{dy}{dx} = \frac{-1}{2}(x^2-5)^{-3/2} \times 2x$$

$$\Rightarrow \frac{dy}{dx} = \frac{-x}{(x^2-5)^{3/2}} \quad \text{now at } x = 3,\ dy/dx = -3/8$$

Thus our required line is $y = \frac{-3}{8}x + c$ but it must pass through $(3, 1/2)$, thus for $x = 3$, $y = 1/2 \Rightarrow \frac{1}{2} = \frac{-3}{8} \times 3 + c \Rightarrow c = \frac{13}{8}$

Thus required equation of tangent line: $y = \frac{-3}{8}x + \frac{13}{8}$ Ans.

YOUTUBE.COM/@THEPHYSICSSHELTER

DIFFERENTIATION BY TAKING LOG: Sometimes it becomes easier in problems to take logarithm before differentiation, logarithm converts multiplications and divisions in to sums and differences.

EXAMPLE: If $y = \frac{x^2-1}{x^2+1}$ find $\frac{dy}{dx}$ at $(0,-1)$

Solution: $\ln y = \ln(x^2-1) - \ln(x^2+1)$, Now differentiating wrt x:

$$\frac{1}{y}\frac{dy}{dx} = \frac{1}{x^2-1} \times 2x - \frac{1}{x^2+1} \cdot 2x = 0 \text{ at } x = 0$$

Thus $\frac{dy}{dx} = 0$

EXAMPLE: If $y = x^x$ find dy/dx.

Solution: $\ln y = x \ln x \Rightarrow \frac{1}{y}\frac{dy}{dx} = x \times \frac{1}{x} + \ln(x) = 1 + \ln x$

$$\Rightarrow \frac{dy}{dx} = y(1+\ln x) = x^x(1+\ln(x)) \text{ Ans.}$$

EXAMPLE: If $y = x^y$ find dy/dx at $(1,1)$

Solution: $\ln y = y \ln x \Rightarrow \frac{1}{y}\frac{dy}{dx} = y \times \frac{1}{x} + \ln(x) \cdot \frac{dy}{dx}$

$$\Rightarrow \frac{dy}{dx}\left[\frac{1}{y} - \ln(x)\right] = y/x \Rightarrow \frac{dy}{dx} = \frac{y}{x}\left[\frac{1}{y} - \ln(x)\right]^{-1} = 1 \text{ Ans.}$$

IMPLICIT DIFFERENTIATION: When we differentiate an equation in x and y without clearly expressing y in terms of x, we call it implicit differentiation.

EXAMPLE: If $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ find $\frac{d^2y}{dx^2}$ at $(0,b)$

Solution: $\frac{2x}{a^2} + \frac{2y}{b^2}y' = 0 \Rightarrow$ at $(0,b)$, $y' = 0$

Differentiating once more: $\frac{2}{a^2} + \frac{2}{b^2}[yy'' + y'^2] = 0$

Now putting $y' = 0$ and $y = b$:

YOUTUBE.COM/@THEPHYSICSSHELTER

$$\frac{2}{a^2} + \frac{2}{b^2}\left[b y'' + 0\right] = 0 \Rightarrow y'' = \frac{-b}{a^2} \text{ Ans.}$$

EXAMPLE: Find dy/dx if $x + y = \sin(x - y)$

Solution: $1 + \frac{dy}{dx} = \cos(x-y)\left[1 - \frac{dy}{dx}\right] \Rightarrow \frac{dy}{dx}\left[1 + \cos(x-y)\right] = \cos(x-y) - 1$

$$\Rightarrow \frac{dy}{dx} = \frac{\cos(x-y) - 1}{\cos(x-y) + 1} \text{ Ans.}$$

PARAMETRIC DIFFERENTIATION: Sometimes y may not be given as a function of x but both y and x may be given as a function of t then:

$$\frac{dy}{dx} = \frac{dy}{dt} \Big/ \frac{dx}{dt}$$

For example if $y = \cos(t)$ and $x = \sqrt{t^2+1}$ then:

$\frac{dy}{dt} = -\sin(t)$ and $\frac{dx}{dt} = \frac{1}{2\sqrt{t^2+1}} \times 2t = \frac{t}{\sqrt{t^2+1}}$, thus:

$$\frac{dy}{dx} = \frac{\sqrt{t^2+1}}{t}\sin(t) \text{ Ans.}$$

EXAMPLE: If $y = e^x \tan x + x \ln(x)$ find dy/dt.

Solution: $\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$, we do not know $x(t)$ so we will just leave dx/dt part as it is.

Now $\frac{dy}{dx} = e^x \cdot \sec^2 x + \tan x\, e^x + x \times \frac{1}{x} + \ln(x)$

$$\Rightarrow \frac{dy}{dt} = \left[e^x\left[\tan x + \sec^2(x)\right) + 1 + \ln(x)\right]\frac{dx}{dt} \text{ Ans.}$$

YOUTUBE.COM/@THEPHYSICSSHELTER

EXAMPLE: If $y = \frac{1}{x}\ln x + e^x \sin 2x + \log_5 x$ find $dy/d\theta$

Solution: $dy/d\theta = \frac{dy}{dx} \times \frac{dx}{d\theta}$., Also $\log_5 x = \frac{\ln x}{\ln(5)}$

(Standard result in logarithms : $\log_a b = \log_c b / \log_c a$)
e.g. $\log_8 64 = \log_2 64 / \log_2 8$)

Now differentiating wrt x:

$$\frac{dy}{dx} = \frac{1}{x} \times \frac{1}{x} + \ln x \cdot \left(\frac{-1}{x^2}\right) + e^x \cdot 2\cos 2x + e^x \sin 2x$$

$$+ \frac{1}{\ln(5)} \cdot \frac{1}{x}$$, Now on simplifying, we get the

required Answer $= \left(\frac{1-\ln x}{x^2} + e^x[\sin 2x + 2\cos 2x] + \frac{1}{x\ln 5}\right) \cdot \frac{dx}{d\theta}$ Ans.

→ Remember to put $dx/d\theta$.

EXAMPLE: If $y = \ln(\tan^{-1}\sqrt{1+x^2})$ find dy/dt.

Solution: $\frac{dy}{dx} = \frac{1}{\tan^{-1}\sqrt{1+x^2}} \cdot \frac{1}{1+(1+x^2)} \cdot \frac{1}{2\sqrt{1+x^2}} \cdot 2x$

Thus required answer:

$$\frac{dy}{dt} = \frac{x}{\tan^{-1}\sqrt{1+x^2}\,(2+x^2)\sqrt{1+x^2}} \cdot \frac{dx}{dt}$$ Ans.

EXAMPLE: If $y = (\sin x)^{\ln x}$, find dy/dx using logarithmic differentiation.

Solution: $\ln y = \ln(x)\ln\sin(x) \Rightarrow \frac{1}{y}\frac{dy}{dx} = \ln(x) \cdot \frac{1}{\sin(x)} \cdot \cos(x) + \ln\sin(x) \cdot \frac{1}{x}$

$$\Rightarrow \frac{dy}{dx} = (\sin x)^{\ln x}\left[\cot x \ln(x) + \frac{1}{x}\ln\sin(x)\right]$$ Ans.

→ For any function of the form $f(x)^{g(x)}$ generally one should take logarithm before differentiating to avoid any fallacies. This form is not a simple nested function. x is an argument of both $f(x)$ and $g(x)$, so direct application of chain rule can't be done. Chain rule is applied for nested functions.

AMBARISH SIR

YOUTUBE.COM/@THEPHYSICSSHELTER